

THE TWO-SUBALGEBRA SHARPNESS CASE FOR A MARGINAL-SUBSPACE CONJECTURE

AGENT_LANE_04

SOURCE AND TARGET

The source is Ivan Feshchenko, *When is the sum of complemented subspaces complemented?*, arXiv:1606.08048. In Section 3.4, after proving that $r(E) < 1$ is sufficient for the centered marginal subspaces

$$L_0^p(\mathcal{F}_1) + \cdots + L_0^p(\mathcal{F}_n)$$

to be complemented, the paper asks whether this condition is sharp and states a conjecture: if $E = (e_{ij})$ is symmetric, has zero diagonal, nonnegative off-diagonal entries, and $r(E) = 1$, then there should be a probability space and sub- σ -algebras $\mathcal{F}_1, \dots, \mathcal{F}_n$ with

$$\psi'(\mathcal{F}_i, \mathcal{F}_j) = 1 - e_{ij}$$

such that the centered marginal sum is not closed in L^p for arbitrary $p \in [1, \infty]$.

This packet proves the conjecture for $n = 2$. For $n = 2$, the only such matrix is

$$E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

DEFINITIONS

For sub- σ -algebras $\mathcal{A}, \mathcal{B}$ of a probability space $(\Omega, \mathcal{F}, \mu)$, Feshchenko uses

$$\psi'(\mathcal{A}, \mathcal{B}) = \inf \left\{ \frac{\mu(A \cap B)}{\mu(A)\mu(B)} : A \in \mathcal{A}, B \in \mathcal{B}, \mu(A) > 0, \mu(B) > 0 \right\}.$$

Also $L_0^p(\mathcal{A})$ denotes the $\mathcal{A}$ -measurable L^p -functions of mean zero.

MAIN RESULT

Theorem 1. *There is a probability space $(\Omega, \mathcal{F}, \mu)$ and two sub- σ -algebras $\mathcal{A}, \mathcal{B} \subset \mathcal{F}$ such that*

$$\psi'(\mathcal{A}, \mathcal{B}) = 0$$

and, for every $p \in [1, \infty]$,

$$L_0^p(\mathcal{A}) + L_0^p(\mathcal{B})$$

is not closed in $L^p(\Omega, \mathcal{F}, \mu)$. Consequently Feshchenko's Section 3.4 conjecture is true for $n = 2$.

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PROOF INTUITION

The sample space is the edge set of an infinite bipartite path. Row-measurable functions are potentials on one side of the path, and column-measurable functions are potentials on the other side. A row potential a_i and a column potential $-a_j$ cancel exactly on the diagonal edges $(i, j) = (m, m)$ and cancel up to the increment $a_{m+1} - a_m$ on the adjacent edges $(m + 1, m)$. Long, slowly varying triangular potentials therefore give row/column functions whose sum is very small although neither centered function is small. This rules out closedness of the sum.

CONSTRUCTION

Choose odd integers $L_k \geq 5$ with $L_k \rightarrow \infty$, and partition $\mathbb{N}$ into consecutive intervals

$$J_k = \{s_k, s_k + 1, \dots, s_k + L_k - 1\}.$$

Put

$$q_m = \frac{2^{-k}}{L_k} \quad (m \in J_k).$$

Then $\sum_{m \geq 1} q_m = 1$.

Let

$$\Omega = \{a_m, b_m : m \in \mathbb{N}\},$$

where a_m is the edge joining row m to column m , and b_m is the edge joining row $m + 1$ to column m . Define

$$\mu(\{a_m\}) = \mu(\{b_m\}) = \frac{q_m}{2}.$$

Let $\mathcal{A}$ be the σ -algebra generated by the row map

$$\rho(a_m) = m, \quad \rho(b_m) = m + 1,$$

and let $\mathcal{B}$ be the σ -algebra generated by the column map

$$\kappa(a_m) = \kappa(b_m) = m.$$

Lemma 1. $\psi'(\mathcal{A}, \mathcal{B}) = 0$, and $L_0^p(\mathcal{A}) \cap L_0^p(\mathcal{B}) = \{0\}$ for every $p \in [1, \infty]$.

Proof. For instance, the row atom $\rho^{-1}(\{1\})$ and the column atom $\kappa^{-1}(\{3\})$ have positive measure and empty intersection. Hence $\psi'(\mathcal{A}, \mathcal{B}) = 0$.

If an $\mathcal{A}$ -measurable function f and a $\mathcal{B}$ -measurable function g agree almost everywhere, write their row and column values as α_i and β_j . Since every edge has positive measure, the edge a_m gives $\alpha_m = \beta_m$, while the edge b_m gives $\alpha_{m+1} = \beta_m$. Thus

$$\alpha_m = \beta_m = \alpha_{m+1}$$

for every m , so all row and column values are equal to one constant. If the common function has mean zero, that constant is zero. $\square$

FAILURE OF CLOSEDNESS

For $m \in J_k$, let c_k be the midpoint of J_k and put

$$\theta_k(m) = 1 - \frac{|m - c_k|}{(L_k - 1)/2};$$

put $\theta_k(m) = 0$ for $m \notin J_k$. Then

$$0 \leq \theta_k \leq 1, \quad \max_m \theta_k(m) = 1, \quad |\theta_k(m+1) - \theta_k(m)| \leq \frac{2}{L_k - 1}.$$

Define bounded functions

$$u_k(\omega) = \theta_k(\rho(\omega)), \quad v_k(\omega) = -\theta_k(\kappa(\omega)).$$

Thus u_k is $\mathcal{A}$ -measurable and v_k is $\mathcal{B}$ -measurable. On a_m one has $u_k + v_k = 0$, and on b_m one has

$$(u_k + v_k)(b_m) = \theta_k(m+1) - \theta_k(m).$$

Hence

$$(1) \quad \|u_k + v_k\|_\infty \leq \frac{2}{L_k - 1}.$$

Let

$$U_k = u_k - \int u_k d\mu, \quad V_k = v_k - \int v_k d\mu.$$

Then $U_k \in L_0^p(\mathcal{A})$ and $V_k \in L_0^p(\mathcal{B})$ for every p . Since

$$U_k + V_k = (u_k + v_k) - \int (u_k + v_k) d\mu,$$

(1) gives

$$(2) \quad \|U_k + V_k\|_\infty \leq \frac{4}{L_k - 1}.$$

For finite p , let $W_k = 2^{-k}$, the total mass of the columns in J_k . A fixed positive fraction of the column atoms in J_k have $\theta_k \geq 1/2$, while $\int |v_k| d\mu \leq W_k \rightarrow 0$. Therefore, for each $p < \infty$, there is a constant $c_p > 0$, independent of k , such that for all large k

$$\|V_k\|_p \geq c_p W_k^{1/p}.$$

The same estimate holds for U_k , after ignoring at most the two endpoint row atoms. On the other hand, (1) and the fact that $u_k + v_k$ is supported on edges with $m \in J_k$ give a constant C_p such that

$$\|U_k + V_k\|_p \leq C_p \frac{W_k^{1/p}}{L_k}.$$

Thus

$$(3) \quad \frac{\|U_k + V_k\|_p}{\|U_k\|_p + \|V_k\|_p} \rightarrow 0 \quad (1 \leq p < \infty).$$

For $p = \infty$, the functions θ_k have oscillation 1, whereas $\int u_k d\mu \rightarrow 0$ and $\int v_k d\mu \rightarrow 0$. Hence

$$\|U_k\|_\infty + \|V_k\|_\infty \geq 1$$

for all large k , while (2) tends to zero. Therefore (3) also holds for $p = \infty$, with the usual interpretation.

Now fix $p \in [1, \infty]$ and set

$$M = L_0^p(\mathcal{A}), \quad N = L_0^p(\mathcal{B}).$$

The addition map

$$T : M \oplus_1 N \rightarrow L^p(\Omega), \quad T(f, g) = f + g,$$

is bounded and injective by the lemma. If $M + N$ were closed, then T would be a Banach-space isomorphism from $M \oplus_1 N$ onto $M + N$, and its inverse would be bounded. Equivalently, there would exist $\delta > 0$ with

$$\|f + g\|_p \geq \delta(\|f\|_p + \|g\|_p) \quad (f \in M, g \in N).$$

The functions U_k, V_k violate this estimate by (3). Therefore $M + N$ is not closed.

Since $E = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ has spectral radius 1 and $\psi'(\mathcal{A}, \mathcal{B}) = 0 = 1 - 1$, this proves the $n = 2$ case of the conjecture.

VERIFICATION AND LIMITATIONS

The proof is self-contained after the source definitions. It proves only the two-subalgebra case. The full n -subalgebra conjecture remains open here, especially for matrices of spectral radius one with all off-diagonal entries strictly below one, such as the 3×3 matrix with all off-diagonal entries $1/2$.

Bounded novelty search was performed on 2026-06-29 using the source title and phrases involving ψ' , marginal subspaces, and nonclosed sums. No later paper resolving this exact $n = 2$ sharpness case was found in the search results.

REFERENCES

- [1] I. Feshchenko, *When is the sum of complemented subspaces complemented?*, arXiv:1606.08048.